RECORD 11

SYSTEMIC DATASTORE CORRUPTION AND INFRASTRUCTURE COMPLIANCE

Engine Architecture Incident Report | Documented by Hamgram77 | October 17, 2026

Overview: This record documents the operational friction that can arise when creators report widespread DataStore or Cloud API failures but initial triage treats the incident as an isolated scripting defect. It separates platform-incident indicators from creator-code defects and proposes a shared evidence, communication, and recovery framework.

1. INCIDENT DEFINITION AND SCOPE

A DataStore incident can range from elevated latency and transient request failures to partial writes, stale reads, or lost progression. Not every error indicates platform failure: request-budget exhaustion, incorrect retry logic, non-idempotent writes, or unsafe schema changes can originate in creator code. The core compliance problem is the absence of a fast, evidence-based method to distinguish these categories during a widespread event.

- Systems in scope: DataStore request handling, related Cloud API paths, status communication, support triage, and recovery guidance.
- Potential impact: player progression, inventories, entitlements, studio reputation, and support workload.

2. FAILURE-PATH MODEL

Layer	Creator-visible symptom	Question for triage
Game logic	Errors cluster around one code path or release.	Did a recent deployment change keys, schemas, retries, or request volume?
Request budget	Throttling rises under load or repeated retries.	Are requests within documented budgets and using backoff?
Platform API	Multiple experiences report similar 5xx errors or latency.	Is there a correlated service incident or regional degradation?
Persistence	Writes appear successful but later reads are stale or missing.	Are versions, timestamps, and update semantics consistent?
Player recovery	Progress or paid entitlements appear missing after the event.	What authoritative snapshot or reconciliation process exists?

3. CORE INFRASTRUCTURE ISSUES

3.1 Deflective triage

When widespread HTTP 500-series errors appear after a platform change, a default 'user scripting bug' response can delay incident correlation. A good first response should still ask creators to check budgets and code changes, but it should also compare timestamps, regions, endpoints, and reports from unrelated experiences.

3.2 Burden of proof

Creators may be asked for exhaustive HAR files or full network captures before a platform incident is acknowledged. Those files can contain sensitive tokens or account details. A safer process would accept a minimal structured incident bundle first, then request targeted diagnostics only when necessary through an authenticated channel.

3.3 Partial failure and corruption risk

Retries can amplify an outage if operations are not idempotent. A timeout also does not always prove that a write failed; blindly repeating a transaction can create duplication or overwrite newer state. Platform guidance should distinguish unavailable, throttled, timed-out, and conflict outcomes.

3.4 Player-community fallout

When players lose progression or paid entitlements, the studio becomes the visible point of contact. Without an official incident identifier or recovery template, creators must explain an event they cannot independently confirm, increasing distrust and inconsistent compensation.

4. MINIMUM INCIDENT EVIDENCE BUNDLE

- UTC timestamps for the first error, peak failure window, and recovery attempts.
- UniverseId or PlaceId, affected environment, approximate server count, and geographic pattern when known.
- Error class and sanitized message, including HTTP status or platform error code without authentication data.
- A short reproduction description and confirmation of whether a recent deployment occurred.
- Request-rate and budget observations, retry/backoff behavior, and whether failures affect unrelated keys or users.
- A redacted sample of correlation or request identifiers, submitted only through an official secure support route.

5. TRIAGE DECISION MATRIX

Signal	Leans creator-side	Leans platform-wide
Timing	Begins immediately after one studio release	Begins without a release and aligns across unrelated games
Scope	One key family, code path, or server cohort	Multiple endpoints, regions, or independent experiences
Error type	Validation, schema, or deterministic application error	Elevated 5xx responses, timeouts, or service-unavailable states
Rollback result	Reverting the release restores behavior	Errors persist on the prior known-good build
Status evidence	No broader reports or status change	Official incident notice or correlated creator reports

6. CREATOR-SIDE RESILIENCE CONTROLS

These practices reduce damage but cannot eliminate platform-side failures. They should be validated against current official documentation before deployment.

- Use bounded retries with exponential backoff and jitter; avoid infinite retry storms.
- Design critical writes to be idempotent and attach operation identifiers where the architecture permits.
- Prefer conflict-aware update patterns for shared state rather than blind overwrite behavior.
- Maintain schema versions, migration safeguards, and recovery checkpoints for high-value progression.

- Separate player-facing recovery messaging from speculative root-cause claims.
- Test failure handling in a controlled environment and document the studio's rollback and compensation authority.

7. REQUIRED PLATFORM REFORMS

- Publish real-time status indicators for DataStore latency, errors, throttling, and regional degradation.
- Issue stable incident identifiers that support teams, creators, and status dashboards can reference consistently.
- Provide safe diagnostic templates that request only the minimum data required for correlation.
- Define when engineering will initiate rollback, traffic reduction, or recovery procedures.
- Publish recovery guidance for stale reads, uncertain writes, duplicated operations, and entitlement reconciliation.
- Provide post-incident summaries with affected windows, impact categories, and creator action items.

8. PLAYER COMMUNICATION TEMPLATE REQUIREMENTS

During a confirmed platform incident, creators need a neutral message that acknowledges service instability, discourages repeated purchases or risky retries, explains that data is being assessed, and gives a time for the next update. The message should not promise restoration until the authoritative state and recovery plan are known.

9. CONCLUSION

Infrastructure compliance is a shared responsibility. Creators must build defensive persistence logic, while the platform must provide observable service health, proportional evidence requests, incident correlation, and recovery guidance. Transparent status metrics and stable incident identifiers would reduce blame-shifting and shorten time to containment.

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